

Basic Biosolids Calculations

This is intended as an aid for performing basic biosolids loading calculations for record keeping and reporting purposes.
This worksheet does not relieve the permittee from the responsibility for performing accurate calculations.

1. DETERMINE DRY TONS

Enter Percent Total Solids (%TS):	2.62
Enter Specific Bulk Density (lb/cubic yard) ¹ :	0
¹ (if quantity is in cubic yards)	
	Quantity Units
Enter the biosolids quantity and units:	87600 gallons
Results:	Dry Tons: 9.5706504
	Dry Metric Tons (for EPA): 8.6823476119

2. DETERMINE DRY TONS PER ACRE

Enter Dry Tons:	9.5706504
Enter Acres:	8
Results:	Dry Tons Per Acre: 1.1963313
	Dry Metric Tons Per Hectare (for EPA): 2.6818075376

3. DETERMINE NUTRIENT LOADINGS

Enter Dry Tons Per Acre:	1.1963313
Enter Percent Total Nitrogen (%TN) ² :	6.41
Enter Percent Total Phosphorus (%TP) ² :	5.76
Enter Percent Total Potassium (%TK) ² :	0.21

²Use this to convert from mg/kg to % if needed

	mg/kg =	0	%
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Results:

lb/acre TN:	153.36967266
lb/acre TP:	137.81736576
lb/acre TK:	5.02459146

lb/acre PAN ³	102.24644844
lb/acre P ₂ O ₅	316.97994125

³ PAN based on 1.5 DEP factor. Calculations for PAN using EPA method are not provided.

kg/ha TN (for EPA):	171.903882106
kg/ha TP (for EPA):	154.472131191
kg/ha TK (for EPA):	5.6317964497

4. DETERMINE METALS LOADINGS

Enter Dry Tons Per Acre:	1.1963313
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Enter Metals Concentrations (mg/kg):

		Loading Results:	
Metal	mg/kg	Metal	lb/acre
Arsenic	17.6	As	0.0421108618
Cadmium	4.1	Cd	0.0098099167
Copper	866	Cu	2.0720458116
Lead	21.9	Pb	0.0523993109
Mercury	4	Hg	0.0095706504
Molybdenum	26.7	Mo	0.0638840914
Nickel	30.2	Ni	0.0722584105
Selenium	5.7	Se	0.0136381768
Zinc	1630	Zn	3.900040038

kg/ha (for EPA)
0.0471998179
0.0109954121
2.3224455835
0.0587315915
0.0107272313
0.0716042691
0.0809905966
0.0152863046
4.3713467681